

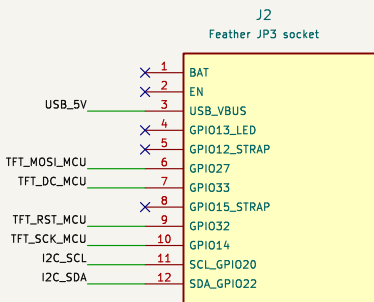
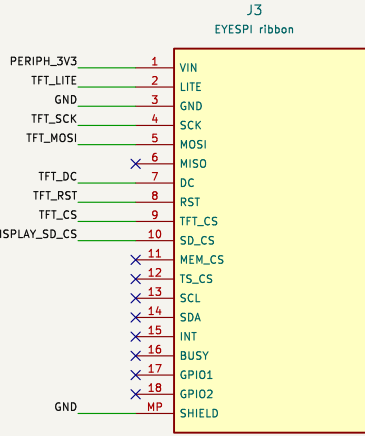
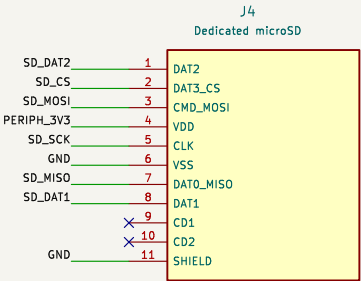
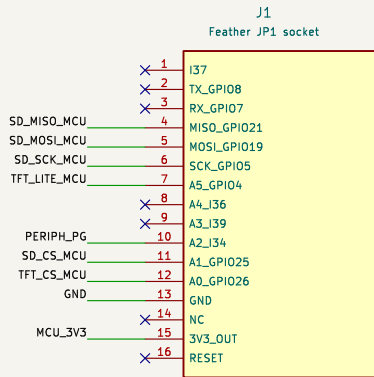
HARMONY MARK-6 | USB-FIRST CARRIER | REV A DRAFT

Review schematic. No battery connected. Separate SPI hosts. All carrier components and sockets to be factory assembled.

FEATHER V2 #5900
Numbering follows Adafruit vendor JP1 / JP3.

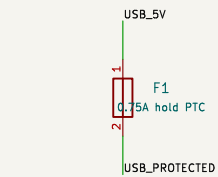
MICROSD | SPI3_HOST

3.5-INCH WAVESHARE 29318 / ST7796S | SPI2_HOST

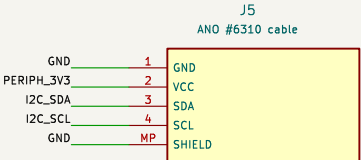
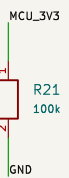
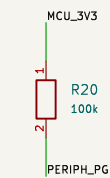
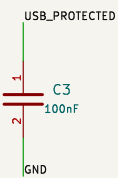
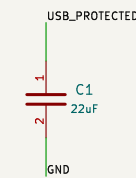
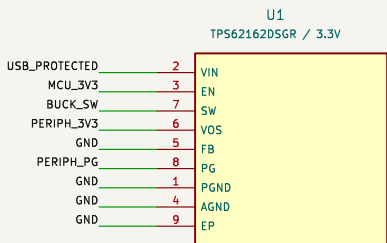


JP1.14 is NOT a power pin. Leave open.
BAT and EN are not connected on this USB-first carrier.
GPIO5 keeps its normal high boot strap; SD CLK is an input.

USB POWER | SEPARATE PERIPHERAL REGULATOR



Pull up CMD and DAT0..3 in SPI mode.
C4/C5 close to the socket; shield to ground.
Initial SD clock: 4 MHz. Card-specific transients need measurement.



ANO has onboard I2C pullups. Poll buttons/encoder.
Use a QT-to-QT cable; not the loose-wire cable.
Do not connect the Feather QT power rail in parallel.

Display SD slot stays empty and deselected. Touch unused.
Display uses SPI; no jumper soldering required.
YESPI VIN = 3.3 V; leave display 15-pin connector unused.
Initial display clock: 4 MHz; bounded refresh transfers.
FPC: 0.5mm pitch, 18-way, 0.3mm, A-B pin1-to-pin1.

HARMONY-17 - schematic review only, not for manufacture

Sheet: /
File: harmony-mark6-carrier.kicad_sch

Title: Harmony Mark-6 USB carrier

Size: A2 Date: 2026-09-21

KiCad E.D.A. B.0.9

Rev: A-DRAFT

Id: 1/1